

Question B2

This question carries 18 marks in total.

(a) [3 MARKS] Find a scalar potential for the following vector field:

$$\mathbf{F}(x, y, z) = (2x + ze^{xz}, z, y + xe^{xz})$$

(b) [3 MARKS] Let $f(x, y, z) = x + ze^{-xy} - 1 = 0$ define a surface. Find the general form for the slope of the surface in the direction of the vector $\mathbf{v} = (0, -1, 1)$.

(c) [6 MARKS] Find the angle between the normals to the surfaces

$$\sin(x) + \cos(y) - \frac{xy}{\pi} + z = 2 \quad \text{and} \quad z = x^2 - (y - \pi)^2 + 1$$

at the point $(\pi, 0, 1)$.

(d) [3 MARKS] Given $\mathbf{F} = yz\mathbf{i} + 5xz\mathbf{j} - (x - y - z)\mathbf{k}$ find $\nabla \times \mathbf{F}$.

(e) [3 MARKS] Calculate the volume of the solid bounded by $z = 2 - x^2 - y^2$ and $z = \sqrt{x^2 + y^2}$.

Solutions

Answer (a):

We seek a scalar potential $\phi(x, y, z)$ such that $\nabla\phi = \mathbf{F}$. This gives us three partial differential equations:

$$1) \frac{\partial\phi}{\partial x} = 2x + ze^{xz}$$

$$2) \frac{\partial\phi}{\partial y} = z$$

$$3) \frac{\partial\phi}{\partial z} = y + xe^{xz}$$

Integrating (1) with respect to x :

$$\phi(x, y, z) = \int (2x + ze^{xz}) dx = x^2 + e^{xz} + g(y, z)$$

Differentiating this result with respect to y and equating to (2):

$$\frac{\partial\phi}{\partial y} = \frac{\partial}{\partial y}(x^2 + e^{xz} + g(y, z)) = \frac{\partial g}{\partial y} = z$$

Integrating with respect to y :

$$g(y, z) = yz + h(z)$$

So now, $\phi(x, y, z) = x^2 + e^{xz} + yz + h(z)$.

Differentiating with respect to z and equating to (3):

$$\frac{\partial \phi}{\partial z} = xe^{xz} + y + h'(z) = y + xe^{xz}$$

This implies $h'(z) = 0$, so $h(z) = C$ (a constant).

Thus, the scalar potential is:

$$\phi(x, y, z) = x^2 + e^{xz} + yz + C$$

Answer (b):

The "slope" of the function $f(x, y, z)$ in the direction of vector $\mathbf{v}$ is its directional derivative $D_{\mathbf{v}}f = \nabla f \cdot \hat{\mathbf{v}}$.

First, calculate the gradient ∇f :

$$f_x = 1 - yze^{-xy}$$

$$f_y = -xze^{-xy}$$

$$f_z = e^{-xy}$$

$$\nabla f = \langle 1 - yze^{-xy}, -xze^{-xy}, e^{-xy} \rangle$$

Next, find the unit vector in the direction of $\mathbf{v} = \langle 0, -1, 1 \rangle$:

$$|\mathbf{v}| = \sqrt{0^2 + (-1)^2 + 1^2} = \sqrt{2}$$

$$\hat{\mathbf{v}} = \left\langle 0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$$

Now, calculate the directional derivative:

$$D_{\mathbf{v}}f = \nabla f \cdot \hat{\mathbf{v}} = (1 - yze^{-xy})(0) + (-xze^{-xy})\left(-\frac{1}{\sqrt{2}}\right) + (e^{-xy})\left(\frac{1}{\sqrt{2}}\right)$$

$$D_{\mathbf{v}}f = \frac{1}{\sqrt{2}}(xze^{-xy} + e^{-xy}) = \frac{e^{-xy}}{\sqrt{2}}(xz + 1)$$

Answer (c):

Let the two surfaces be defined by level curves $F(x, y, z) = 0$ and $G(x, y, z) = 0$:

$$F(x, y, z) = \sin(x) + \cos(y) - \frac{xy}{\pi} + z - 2 = 0$$

$$G(x, y, z) = z - x^2 + (y - \pi)^2 - 1 = 0$$

The normals to the surfaces are given by their gradients.

Gradient of F :

$$\nabla F = \left\langle \cos(x) - \frac{y}{\pi}, -\sin(y) - \frac{x}{\pi}, 1 \right\rangle$$

At the point $(\pi, 0, 1)$:

$$\nabla F(\pi, 0, 1) = \left\langle \cos(\pi) - 0, -\sin(0) - \frac{\pi}{\pi}, 1 \right\rangle = \langle -1, -1, 1 \rangle$$

Gradient of G :

$$\nabla G = \langle -2x, 2(y - \pi), 1 \rangle$$

At the point $(\pi, 0, 1)$:

$$\nabla G(\pi, 0, 1) = \langle -2\pi, 2(0 - \pi), 1 \rangle = \langle -2\pi, -2\pi, 1 \rangle$$

The angle θ between the normals satisfies $\cos \theta = \frac{\nabla F \cdot \nabla G}{|\nabla F| |\nabla G|}$:

$$\nabla F \cdot \nabla G = (-1)(-2\pi) + (-1)(-2\pi) + (1)(1) = 4\pi + 1$$

$$|\nabla F| = \sqrt{(-1)^2 + (-1)^2 + 1^2} = \sqrt{3}$$

$$|\nabla G| = \sqrt{(-2\pi)^2 + (-2\pi)^2 + 1^2} = \sqrt{8\pi^2 + 1}$$

$$\cos \theta = \frac{4\pi + 1}{\sqrt{3}\sqrt{8\pi^2 + 1}}$$

$$\theta = \arccos \left(\frac{4\pi + 1}{\sqrt{24\pi^2 + 3}} \right)$$

Answer (d):

Given $\mathbf{F} = \langle yz, 5xz, -x + y + z \rangle$, the curl is:

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & 5xz & -x + y + z \end{vmatrix}$$

Calculating the components:

$$\mathbf{i}\text{-component: } \frac{\partial}{\partial y}(-x + y + z) - \frac{\partial}{\partial z}(5xz) = 1 - 5x$$

$$\mathbf{j}\text{-component: } - \left(\frac{\partial}{\partial x}(-x + y + z) - \frac{\partial}{\partial z}(yz) \right) = -(-1 - y) = 1 + y$$

$$\mathbf{k}\text{-component: } \frac{\partial}{\partial x}(5xz) - \frac{\partial}{\partial y}(yz) = 5z - z = 4z$$

So,

$$\nabla \times \mathbf{F} = (1 - 5x)\mathbf{i} + (1 + y)\mathbf{j} + 4z\mathbf{k}$$

Answer (e):

The solid is bounded above by the paraboloid $z = 2 - x^2 - y^2$ and below by the cone $z = \sqrt{x^2 + y^2}$.

We use cylindrical coordinates, where $x^2 + y^2 = r^2$. The surfaces become:

Upper bound: $z = 2 - r^2$

Lower bound: $z = r$

First, find the curve of intersection to determine the limits for r :

$$2 - r^2 = r \implies r^2 + r - 2 = 0 \implies (r + 2)(r - 1) = 0$$

Since $r \geq 0$, the intersection occurs at $r = 1$.

The volume V is given by the triple integral:

$$V = \int_0^{2\pi} \int_0^1 \int_r^{2-r^2} r \, dz \, dr \, d\theta$$

Integrate with respect to z :

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^1 r(2 - r^2 - r) \, dr \, d\theta \\ &= \int_0^{2\pi} d\theta \int_0^1 (2r - r^3 - r^2) \, dr \\ &= 2\pi \left[r^2 - \frac{r^4}{4} - \frac{r^3}{3} \right]_0^1 \\ &= 2\pi \left(1 - \frac{1}{4} - \frac{1}{3} \right) = 2\pi \left(\frac{12 - 3 - 4}{12} \right) = 2\pi \left(\frac{5}{12} \right) \\ V &= \frac{5\pi}{6} \end{aligned}$$